BioTRACE: Mitigating Evaluation Leakage for Knowledge-Chain Extraction from Genomics Scholarly Tables via Source-Isolated Benchmarking

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Abstract

Scholarly tables in genomics and related scientific literature encode valuable conditions, entities, and quantitative measurements for automated knowledge acquisition. However, evaluation can overestimate extraction quality when related documents cross data splits, repair procedures use reference-derived signals, or invalid outputs are excluded from scoring. We introduce **BioTRACE** (*Biomedical Table Records Audited for Chain Extraction*), a source-isolated evaluation protocol and benchmarking framework for knowledge-chain extraction from scholarly tables. BioTRACE enforces document-level split isolation, frozen AI-only test references, formal output contracts, and full-denominator strict Chain-F1. References are produced through two DeepSeek-V3.2 annotation passes followed by DeepSeek-V3.2 adjudication, without human gold annotations or human inter-annotator agreement analysis. On genomics-oriented scholarly tables, a supervised Qwen3-8B [27] extractor achieves higher agreement with the frozen reference than the base model and a deterministic rule, while absolute performance remains low and no statistically reliable advantage over whole-table DeepSeek-V3.2 is established. Matched source-only diagnostics do not establish gains from the tested learned repair routes, and a separate PPO diagnostic provides no positive evidence. These findings highlight the importance of source isolation, contract-aware scoring, and transparent failure accounting before making claims about agent-driven repair in scholarly table knowledge acquisition.

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CCS Concepts

• **Information systems** → **Data extraction and integration**; **Information extraction**; • **Computing methodologies** → **Artificial intelligence**.

Keywords

Knowledge acquisition, Scholarly tables, Knowledge-chain extraction, Source-isolated evaluation, Evaluation leakage

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1 Introduction

A large fraction of claims in scholarly literature is encoded in tables rather than prose [2, 4, 13]. A useful record for knowledge acquisition is not a verbalization of a cell: it must bind a single-focus result to the conditions under which it holds and to the entities it mentions, so that later search or knowledge-base construction can audit provenance [11, 16, 25]. We call each such record a *knowledge chain*. One table often yields many parallel chains, and one-shot extraction commonly omits entities, conflates results, or mismatches conditions (Figure 1). The output schema is genomics-oriented: a closed entity vocabulary includes gene symbols, regulators, and experimental methods. We evaluate that condition–result–entity (CRE) schema on genomics scholarly tables from Europe PMC (Section 4).

The difficulty is evaluation as much as extraction. Large language models (LLMs) make end-to-end generation attractive, and iterative self-repair is a common second stage [15, 21]. Reported gains can be optimistic in three distinct ways. First, tables from the same document share headers, footnotes, entity sets, and sometimes identical bodies; a table-level split then measures same-document generalization rather than extraction from a new document [5, 19,

23]. Second, a correction loop can use a reference-derived score to select actions, accept edits, or roll back to the best observed state; that procedure is adaptive test-set optimization, not source-only inference [8, 10]. Third, scoring only contract-valid outputs estimates $E[\text{F1} \mid \text{valid}]$ rather than end-to-end $E[\text{F1}]$, so systems that fail on hard tables and are dropped from the mean can inflate conditional performance.

We introduce **BioTRACE** (*Biomedical Table Records Audited for Chain Extraction*) as a source-isolated evaluation protocol and benchmarking framework. *Audited* means split, contract, and failure accounting, not human factual verification. BioTRACE groups tables by source document, freezes AI-only references before any test prediction, and scores every table in a fixed denominator under a nested output contract (strict Chain-F1). The protocol does not claim a successful extractor; it defines an estimand for source-only, reference-blind, full-denominator agreement with a frozen chain set. Table 2 maps the three leakage threats to the corresponding controls. Figure 2 presents the BioTRACE framework under which those controls are applied.

On a 120-table / 105-source Final Unseen set, a frozen supervised Qwen3-8B extractor [27] has higher observed AI-reference agreement than the untuned public checkpoint and a deterministic rule. Absolute Chain-F1 remains low. Source-clustered intervals establish an advantage only over the Base checkpoint, not over the rule or whole-table DeepSeek-V3.2. Field-level F1 substantially exceeds chain-level F1: recovering condition, result, or entity fields does not imply joint relational correctness. Matched source-only diagnostics do not establish extra utility from the tested learned repair routes, and a PPO diagnostic is not advanced because it fails a development-stage utility gate.

We make three contributions.

(1) Protocol construction. BioTRACE implements controls for source overlap, reference-derived repair, and valid-only scoring (Figure 2, Table 2, Table 1).

(2) Held-out evaluation. Under this protocol, supervised extraction has higher point-estimate agreement than Base and a rule on new sources, while absolute chain quality remains low (Figure 3, Table 6).

(3) Error decomposition. We separate output validity, field recovery, chain binding, and repair utility, showing that binding errors are a distinct and consequential source of the gap between field-level and chain-level scores (Figure 4, Figure 5).

The remainder of the paper is organized as follows. Section 2 situates the protocol among table mining, leakage-aware evaluation, and LLM repair. Section 3 defines the task, threat model, and estimand. Section 4 specifies protocol and benchmark construction. Section 5 answers four research questions. Section 6 discusses implications and limitations.

2 Related Work

Scholarly table extraction. WebTables established tables as a web-scale knowledge resource [2]. Crestan and Pantel [4] and Muñoz et al. [16] established table mining as a WSDM topic. Semantic table interpretation maps cells and columns to knowledge-graph elements [12]; Tab2Know and related pipelines emit column/cell links or n-ary tuples from scientific PDFs, often with human review as a quality gate [11, 17]. S2ORC treats tables as first-class scholarly objects [13]; PubTables-1M targets detection and structure recognition [22]. SciIE/SciREX extract scientific relations at document scale [9, 14]; schema-driven LLM extractors and materials pipelines produce structured records from heterogeneous tables [1, 6, 18]. Table QA and verification (TAPAS, TabFact, Chain-of-Table, TableLlama) answer or check a query against a table [3, 7, 24, 26, 28]. These lines annotate columns, cells, or query answers. They do not score an unordered, contract-valid CRE *set* for a whole table under full-denominator failure accounting.

Leakage-aware evaluation. Contamination and split leakage can invert scientific conclusions [5, 19]. Random or table-level splits overestimate generalization to new sources [23]. SciREX's public history is an illustrative case: dropping table-only relations rewrites the denominator and changes precision [9]. LLM-as-a-judge protocols scale annotation but inherit position, verbosity, and self-enhancement biases [29]. BioTRACE treats overlapping historical tables as a development diagnostic (Table 5) and scores Unseen agreement with an AI-only adjudicated reference.

LLM repair and reference contamination. Self-Refine and Reflexion popularized iterative revision [15, 21]. Intrinsic self-correction without external correctness signals often fails, and some reported gains depend on oracle labels [8, 10]. PPO is a generic policy optimizer [20]. BioTRACE tests whether learned repair retains utility when the reference is frozen and invisible at prediction time; it does not propose a stronger repair agent.

Research gap. Prior work discusses data leakage, LLM-as-judge bias, and self-repair separately. What is missing is a knowledge-chain benchmark that unifies source isolation, reference blindness, and contract failure accounting, and that reports validity, field recovery, and chain binding as distinct layers.

3 Task and Evaluation Threat Model

3.1 Task

Given a serialized scholarly table (cells, headers, caption/footnotes when present in the source string), a system must emit a JSON object containing a `knowledge_mining` array of at most 16 chain records. The cap is part of the task definition: references and predictions are generated under the bounded contract, and an output exceeding 16 records is contract-invalid rather than silently truncated. Scores therefore measure bounded chain-set agreement and do not claim complete coverage of tables requiring more than 16 chains. Each record nests a knowledge-chain object (`condition`, `result`) and an entity-recognition object. Semantically, a `result` is the single primary measured endpoint of that chain; a `condition` holds the covariates that localize it (row identity, stratum, dose, time). Syntactically, a condition is one or more `Column:value` fields joined by "|"; a result is *exactly one* such field. The first ":" splits key from value; "|" splits condition fields. There is no escape convention, so a cell that itself contains ":" or "|" can break a string field. Regular keys and all literal values must be source-supported. For structurally irregular tables, the reserved `Row` (condition) and `Value` (result) keys preserve source literals while standing in for an unrecoverable header; these reserved keys carry no scientific semantics. The entity vocabulary is exactly `gene_symbol`, `regulator`, `target`,

`functional_mechanism`, `species`, `disease`, `organ`, `tissue`, `cell`, `protein`, `compound`, `drug`, `experimental_method`, `phenotype`; the object may be empty. Mixing several unrelated metrics into one result is invalid. Knowledge chains are *not* chain-of-thought traces. Figure 1 shows the intended acquisition unit; Table 1 is authoritative if the illustration flattens JSON.

Formally, for table source x the target is an unordered multiset of chains $Y = \{c_i\}_{i=1}^{m}$, $c_i = (C_i, r_i, E_i)$. The evaluation unit is the *whole table*, not a single field.

3.2 Threat model and controls

Table 2 states the evaluation threats and the corresponding controls. Conventional evaluation measures a different estimand from BioTRACE. Figure 2 shows the BioTRACE construction: source-only table preparation, silver knowledge-chain annotation, SFT-1, and the repair/PPO routes; the repair/PPO branches are diagnostic-only, not a claimed successful end-to-end system.

Source isolation. Tables from one paper share column names, abbreviations, experimental templates, entity inventories, and sometimes identical bodies. Grouping by document changes the target from "a new table from a seen source" to "a table from a new source." This is an estimand correction, not a data-cleaning detail.

Reference-blind repair. If a reference-derived score is used to choose an action, accept an edit, or restore a previous round, the test label enters the inference controller even when the model never reads the answer text. Freezing the reference and scoring offline precludes the specified prediction-time path reference → action → prediction under the logged workflow. Historical closed-loop runs that used such selection are therefore not identifiable as source-only repair gains.

Full-denominator scoring. Scoring only valid JSON estimates $E[\text{F1} \mid \text{output valid}]$. A system can obtain high conditional F1 by emitting legal answers only on easy tables. BioTRACE keeps every table in N and assigns 0 to parse, contract-invalid, empty, and API/transport failures.

3.3 Metric and estimand

Primary metric is *strict Chain-F1*. Canonicalization lowercases and applies Unicode NFKC to field values, normalizes numbers, and sorts condition fields; the first ":" still splits key from value. After canonicalization, predicted and reference chains are multisets. True positives are the size of the multiset intersection. Table-level precision, recall, and F1 use that count; if both sets are empty the table scores 1.0. Final Unseen has no empty reference. The reported score is the macro-averaged table-level F1 over 120 tables. Field-level F1 (condition, result, entity) is analysis only.

Scoring is fail-closed and ordered: (i) if the prediction remains contract-invalid, the table scores 0 and stays in N; (ii) otherwise canonicalize and match. A generation-time post-normalizer may later emit valid JSON; that generation success rate is not the scored validity rate (SFT-1 generates parseable JSON on 120/120 vs. fail-closed 100/120). We separately count API/transport failures (zero on Final Unseen, including DeepSeek-V3.2). Empty rule outputs are not API failures.

The BioTRACE estimand is therefore

$$\operatorname*{mean}_{x \sim \mathcal{D}_{\text{new-source}}} \text{Chain-F1}\big(f(x), Y(x)\big),$$

with Y frozen and invisible to f, and with invalid $f(x)$ contributing 0. Legacy R-rule on overlapping tables is a historical diagnostic and is never placed in the same comparison column as Chain-F1 (Table 5).

4 Benchmark Construction

4.1 Acquisition, isolation, and freeze order

Final Unseen is collected from Europe PMC open-access full-text XML. We retrieve open-access records with full text and first publication dates from 1 May 2026 to 13 August 2026 (Europe PMC query `OPEN_ACCESS:Y AND HAS_FT:Y` with `FIRST_PDATE:[2026-05-01 TO 2026-08-13]`). The query is a recency and open-access filter on a biomedical literature service. Source identity is the Europe PMC PMCID. Each record stores table markdown (caption and footnotes when present in the XML), a citation string, and a license/access note. Intake forbids reference chains, predictions, and scores. Table 3 records the collection pipeline, isolation audit, and table/reference statistics, all counted from the frozen source and reference records.

Novelty hashes are SHA-256 of whitespace-normalized, case-normalized table markdown. Quality gates require at least two columns, one nonempty data row, at most 30 columns, at most 60,000 characters, complete provenance fields, and tokenizer length strictly below 4,096 tokens under the frozen SFT-1 prompt format (4 tables rejected). Selection is deterministic: keep the shortest valid table per source (sources sorted by PMCID), then fill with the second-shortest table when a source has one, cap two tables per source, stop at 120, and number tables in source-ID order.

Construction order is fixed: (1) build the historical registry; (2) validate intake; (3) freeze the source-only records; (4) build and freeze the AI-only reference; (5) audit that predictions have not started; (6) generate all four Unseen methods without scoring; (7) freeze predictions; (8) open the reference only to the offline scorer.

4.2 AI-only reference

References are DeepSeek-V3.2 annotations, not a human gold standard and not human IAA. The two isolated model calls acting as annotators A and B use the same configuration (temperature 0, 8192 completion tokens) with identical contract text except distinct role tags, followed by same-model adjudication. They are not two independently designed annotators. Mean A–B strict Chain-F1 is 0.494; exact chain-set agreement is 31/120 (25.8%). Field-level A–B F1 is higher (condition 0.743, result 0.695, entity 0.793). Mean chain cardinality is 10.1 (A), 10.9 (B), and 10.5 (adjudicated). A bounded source-only format retry is allowed on contract failure (at most one retry). The adjudicator sees only the table and the two annotations. It must accept A (32 tables), accept B (3), or merge/revise (85). Empty adjudicated outputs are disallowed when either valid candidate is nonempty. Source and reference checksums are frozen in the released artifact; no Unseen system reads the reference at prediction time. The 200-table development split uses a separate source-isolated AI-only reference under the same output schema.

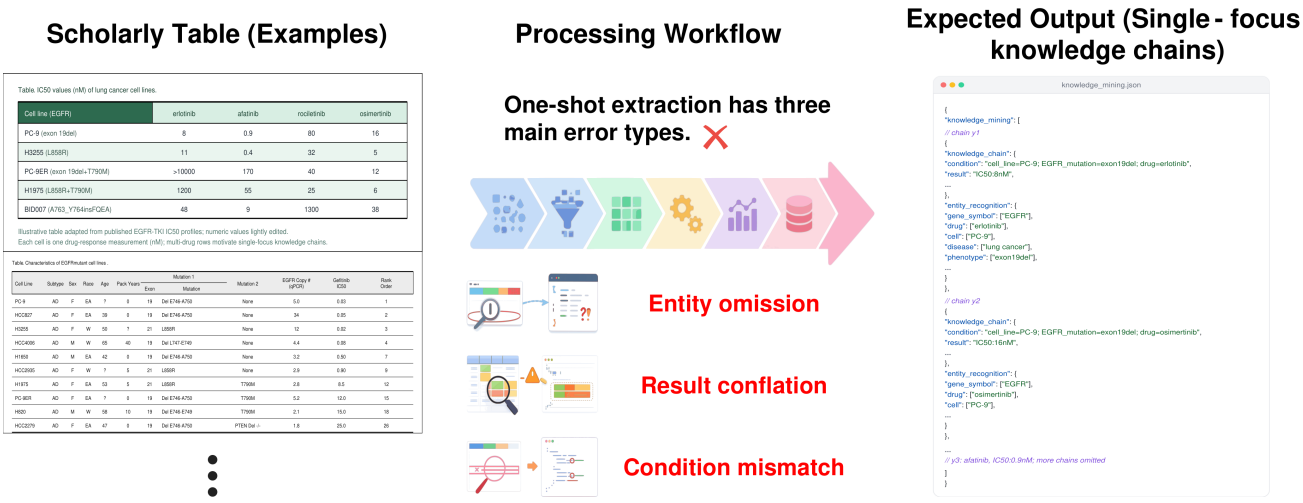


Figure 1: Knowledge-chain extraction from a scholarly table. One table yields many single-focus condition–result–entity records. Typical one-shot errors are entity omission, result conflation, and condition mismatch. The nested contract in Table 1 is authoritative if the figure flattens JSON.

Table 1: Output schema and scoring contract. Prompt text asks for displayed header spelling; regular key/value fields, or the literal payload of the reserved Row/Value fallback, must be source-supported. This is not row/column provenance validation; relational binding is evaluated by the chain matcher against the frozen reference. Parse, empty, and contract-invalid outputs remain in the denominator and score 0.

Component	Required structure	Enforced constraint	Failure
`knowledge_mining`	array in scored system root object only	system root key; length 1–16	missing/non-array → 0
`knowledge_chain`	nested {`condition`, `result`}	nested object present	wrong shape → invalid
`condition/result`	`Column:value` or reserved fallback; one result	parse `key:value`	unparseable / multi-result
Grounding	regular header or fallback literal	normalized source support	ungrounded → invalid
`entity_recognition`	registered types or {}	type vocabulary + intra-chain value	bad type/value
Full prediction	set match vs. frozen Y	strict Chain-F1 on the set	empty/invalid = 0, stays in N

Table 2: BioTRACE threat model. Three leakage channels inflate extraction scores; three controls redefine the estimand as source-only, reference-blind, full-denominator chain agreement. The design does not isolate the effect of any single control.

Threat	Conventional evaluation	BioTRACE control	Targeted risk
Source overlap	table-level or random split; related documents cross train/test	document-level grouping; ID-namespace and table-hash audit	memorization / same-document generalization
Oracle repair	reference-derived scores select, accept, or roll back edits	frozen, prediction-time invisible reference; offline scoring	adaptive test-set optimization
Valid-only scoring	discard parse or contract failures before averaging F1	full-denominator fail-closed Chain-F1	selection bias / $E[\text{F1} \mid \text{valid}]$

4.3 Compared systems

Unseen methods share the source-only table string, the nested contract, a 3,584-token generation budget, and the fail-closed scorer.

Base Qwen. Public Qwen3-8B [27], greedy decoding, no task-specific fine-tuning.

Frozen SFT-1. The same backbone with a frozen LoRA adapter trained as in Section 4.4, then frozen. Greedy decoding, 3584 new tokens, without test-time chain-of-thought.

DeepSeek-V3.2 whole-table. Same model family as isolated calls A/B; one completion at temperature 0, maximum 3584 generated tokens, plus at most one automatic retry of the same prompt if the first output is contract-invalid. On Unseen, 75/120 remain contract-invalid after that retry; API/transport failures are 0.

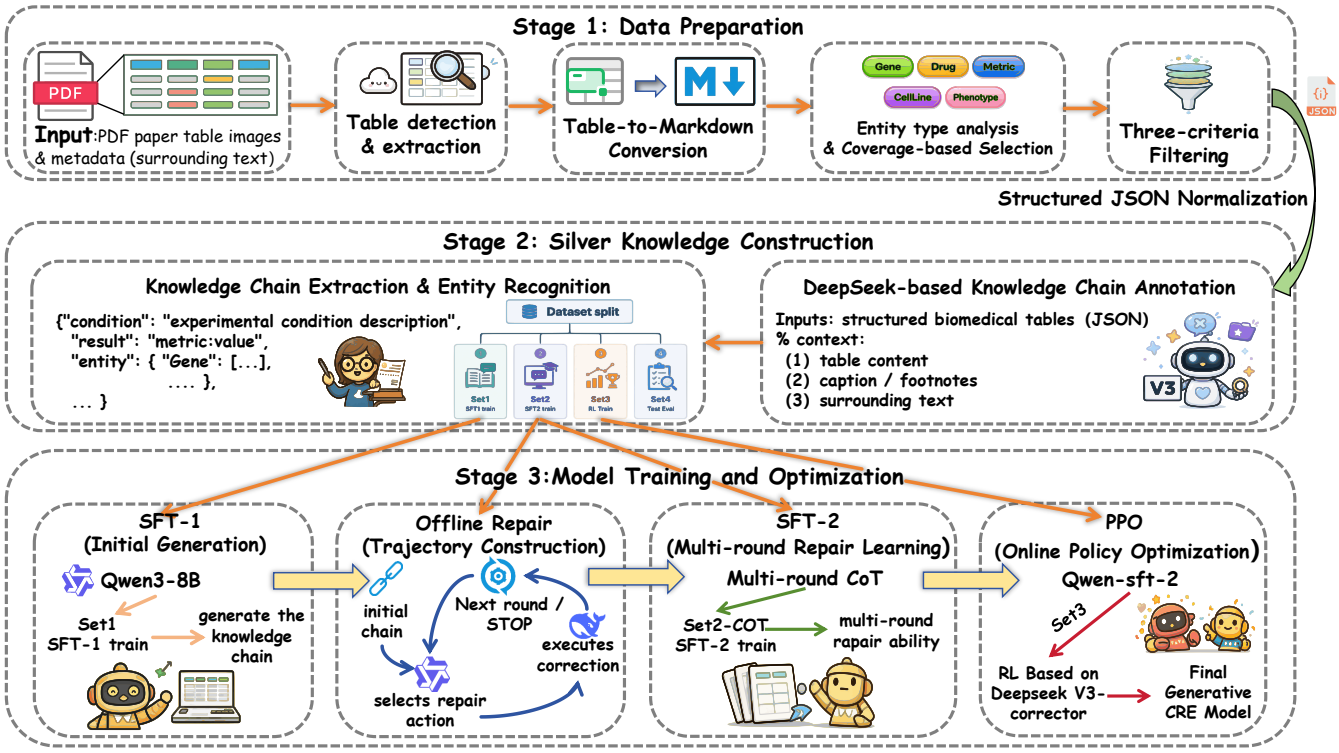


Figure 2: BioTRACE framework. Stage 1 prepares source-only tables. Stage 2 builds silver knowledge chains. Stage 3 trains SFT-1 and optional repair/PPO routes. Final Unseen reports Base, SFT-1, DeepSeek, and the rule; SFT-2 and PPO remain diagnostics under the frozen protocol.

Table 3: BioTRACE Final Unseen dataset card. Counts are from the frozen intake, source, and adjudicated-reference records. Parentheses report first and third quartiles (Q1, Q3).

Item	Value
Intake	500 retrieved records; 106 articles with tables; 180 collected; 176 passed quality filters; 120 selected
Sources	105 documents: 90 contribute one table and 15 contribute two tables (120 tables total)
Isolation	0 overlap vs. 13,156 historical source IDs and 25,166 table hashes
Input tokens	median 870 (Q1 714, Q3 1,147); max 3,723
Table shape	rows median 8 (Q1 5, Q3 15.5); columns median 5 (Q1 3, Q3 7)
Ref. chains	mean 10.5, median 11, range 1–16 (36 tables at cap 16)
Adjudication	accept A 32; accept B 3; merge/revise 85

Deterministic rule. Parse the markdown table; take the first non-empty header/value pair as the condition and each later pair as its own result chain; skip empty, dash, or missing-value cells; cap at 16; if the candidate is contract-invalid, emit an empty chain array.

Repair diagnostics (not Unseen systems). Heuristic or learned (SFT-2) action policies with a fixed external corrector; DeepSeek-V3.2 as a policy; multi-turn recovery; PPO. Actions are Add, Delete, Modify, and STOP. Repair diagnostics are scored offline against a frozen Dev/Train reference; that score is not an inference-time input.

4.4 Training and reproducibility

SFT-1 silver-standard targets are DeepSeek-V3.2 source-only generations under a frozen source-only contract on historical tables, not the Unseen reference and not a human gold standard. Train contains 2,048 unique sample IDs from 1,569 sources and 1,873 unique table hashes (175 duplicate hashes remain). A 100-example development split is a post-training diagnostic only. Table-hash and source-ID overlap with Final Unseen is zero. Table 4 lists the QLoRA hyperparameters. Development-set model selection was disabled, so the released adapter is the final 768-step checkpoint from the single trained seed (42), not a development Chain-F1 pick. We report that checkpoint because the choice was fixed before Final Unseen prediction. Adapter checksums and frozen records are released with the code.

The adapter was frozen before Unseen labels were available; the training log is not a record of a failed run. Final Unseen later designates this adapter as the frozen extractor.

5 Experiments

5.1 Setup

All reported Chain-F1 numbers use the full denominator of the named split. Tables show four decimals; prose uses three unless a

Table 4: SFT-1 training configuration for the frozen Unseen extractor. One seed; final checkpoint (no development-set model selection).

Item	Value
Train / Dev records	2,048 / 100
Sources / unique table hashes	1,569 / 1,873
Epochs / steps / max length	3 / 768 / 4,608
Attention/MLP LoRA	$q/k/v/o$, gate/up/down
LoRA r/α/dropout	16 / 32 / 0.05
LR / batch / accum.	2×10^{-4} / 1 / 8
Optim. / quant.	8-bit AdamW; 4-bit NF4, BF16
Seed / peak GPU memory	42 / 21.95 GiB

confidence interval is quoted. Paired contrasts use 10,000 source-cluster bootstrap replicates (seed 20260813) over 105 source documents, retaining every table from each sampled source. SFT-1 versus Base is the primary contrast; comparisons with the rule and DeepSeek are secondary and descriptive. These intervals are a hash-bound derived audit of the frozen predictions and scorer inputs. Because 15 sources contribute two tables, a replicate can contain a varying number of tables; this preserves the table-macro estimand while clustering dependence by source rather than imposing equal-source weighting. The dated 13 Aug 2026 source protocol fixes the four methods, metric, bootstrap seed, and fail-closed denominator. It is not a public preregistration. No Unseen system reads the reference at prediction time. SFT-2 failed its Dev200 utility gate. The separate PPO diagnostic did not enter Final Unseen; no Unseen score was used for model selection.

5.2 RQ1: How does BioTRACE redefine the evaluation target?

Table 5 assigns evidence roles. The legacy table-level split (N=2,749) has 27.5% exact table-body overlap with training after whitespace-normalized matching, uses a format-oriented R-rule score, and allowed reference-assisted rollback. Under that diagnostic, Base R-rule was 0.4727 and SFT-1 0.6835. Later closed-loop stages rose to 0.6883, and a success-only PPO summary reached 0.6912. These apparently strong development results combine overlap, a format-oriented score, reference-assisted selection, or a success-only denominator. Final Unseen is a different split, a different reference, a different metric, a different failure policy, and a different label-visibility setting. The two evaluations therefore target different estimands. Their discrepancy motivates source-isolated, reference-blind, and fail-closed evaluation; it does *not* identify the isolated effect of any single control, and the two scores are not comparable as a learning curve.

5.3 RQ2: How do extractors perform under the protocol?

Figure 3 and Table 6 report the official adjudicated reference. Base Qwen obtains near-floor Chain-F1 (0.008; 119/120 tables score zero). Frozen SFT-1 reaches 0.159 with fail-closed validity 100/120. The rule is usually contract-valid (112/120) but scores only 0.106 on the chain. DeepSeek scores 0.105 with 75/120 contract-invalid after retry, not transport failures.

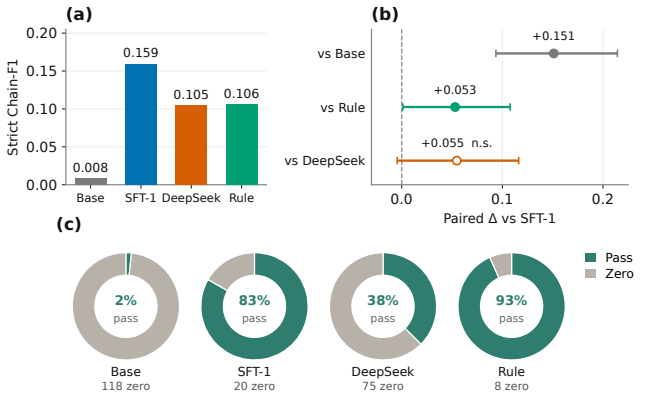


Figure 3: Final Unseen (N=120, 105 sources), strict Chain-F1 vs. the frozen adjudicated AI-only reference. (a) Macro Chain-F1. (b) Paired Δ vs. SFT-1 with source-clustered 95% CI. (c) Contract-valid and contract-invalid table counts.

Figure 3b reports paired source-cluster contrasts. SFT-1 exceeds Base (Δ+0.1511, CI 0.0937–0.2123). SFT-1 versus the rule is Δ+0.0531, CI −0.0058–0.1107, and SFT-1 versus DeepSeek is Δ+0.0548, CI −0.0039–0.1159; both secondary intervals include 0. Scores are frozen-AI-reference agreement, not factual accuracy, including possible same-model-family overlap [29].

Table 7 is a reference-sensitivity analysis: the same frozen predictions rescored against isolated call A, isolated call B, and the adjudicated final set. These are not independent gold standards. SFT-1 remains the highest of the four methods on all three references, and SFT-1 versus Base keeps CIs excluding 0. The SFT-1–Rule interval includes 0 under A and Final, but excludes 0 under B; SFT-1 versus DeepSeek follows the same pattern. DeepSeek versus the rule changes sign (+0.003 on A; −0.006 on B; −0.002 on Final), but every 95% CI includes 0, so the swap is not distinguishable from noise. These comparisons are sensitivity diagnostics, not independent gold standards; the observed extractor-versus-Base/rule pattern remains reference-sensitive, as does the comparison to same-family DeepSeek.

5.4 RQ3: Do errors come from invalid outputs, missing fields, or binding?

Validity, field overlap, and chain identity are three layers (Figure 4). SFT-1 condition/result/entity F1 are 0.386, 0.395, and 0.542, against chain F1 0.159. The rule attains the highest entity F1 (0.617) while losing the chain (0.106). Marginal field accuracy does not imply joint relational correctness: a system can recover the right cells and bind them into the wrong chain.

Table 8 stratifies the same frozen SFT-1 predictions. Reference-chain cardinality does *not* show a decreasing Chain-F1 trend (0.093 on 1–4 chains, n=15; 0.140 on 5–8, n=32; 0.182 on 9–16, n=73, of which 36 reach the cardinality cap of 16). We therefore do not claim that binding errors become more consequential merely as the number of parallel claims increases. In this sample, input length is the largest descriptive difference among the two reported stratifications: Chain-F1 falls from 0.303 on the shortest token tertile to 0.142 then 0.033 on the longest, with exact match 9/40, 3/40,

Table 5: Evidence roles for the numbers in this paper. Legacy R-rule and strict Chain-F1 are different estimands. Final Unseen is the only main result; Dev200 diagnoses repair; Train48 is train-only calibration.

Evidence	N	Source scope	Reference	Metric	Paper role
Legacy split	2,749	hist. overlap 27.5%	assisted	R-rule	development diagnostic
AI-only Dev100	100	source-isolated Dev	AI-only	Chain-F1	extractor diagnostic
AI-only Dev200	200	source-isolated Dev	AI-only	Chain-F1	repair diagnostic
Train48 recovery	48	train-only	AI-only	Chain-F1	calibration, not generalization
Final Unseen	**120 / 105 src.**	new-source, zero overlap	AI-only	Chain-F1	**main result**

Table 6: Official Final Unseen scores (N=120 / 105 sources). Chain-F1 is macro agreement with the frozen adjudicated reference. Fail-closed scoring assigns 0 to any row that remains contract-invalid. Bold marks the best value per column (lowest Invalid).

Method	Chain-F1	Valid	Invalid	Transport
Base Qwen	0.0083	2/120	118/120	0
Frozen SFT-1	**0.1594**	100/120	20/120	0
DeepSeek whole-table	0.1047	45/120	75/120	0
Deterministic rule	0.1064	**112/120**	**8/120**	0

and 0/40. Condition F1 remains higher than chain F1 in every bin. This analysis is descriptive and does not establish a causal effect of length.

Re-scoring the frozen predictions with the official matcher shows that SFT-1 matches the reference exactly on 12/120 tables and scores 0 on 91/120 (20 fail-closed; 71 schema-valid mismatches). On 38 of those valid zeros, condition F1 is already positive. The rule has entity F1 1.0 on 59 tables whose chain F1 is 0. Base is 0 on 119/120; DeepSeek is 0 on 103/120 and exact on 5.

SFT-1's 20 fail-closed zeros are 6 duplicate chains, 10 ungrounded values, 2 unparseable conditions, 1 unsupported key, and 1 disallowed token. DeepSeek-V3.2's 75 contract-invalid rows are dominated by 50 invalid entity values (unregistered types or values that fail the 14-type vocabulary or intra-chain consistency check after one retry), plus 11 grounding failures and 14 other contract errors. Syntactically well-formed JSON is not a schema-valid CRE set. The rule's 8 zeros are all empty outputs.

Table 9 makes the binding failure concrete. On Unseen table 114 (accepted from annotator A), SFT-1 emits the same 14-chain set as the reference (Chain-F1 1.0). The rule is schema-valid and attains entity F1 1.0, but splits each row into per-cell chains, so chain F1 is 0. On Unseen table 002 (merge/revise), SFT-1 and the rule both reach condition and result F1 1.0 yet score chain F1 0, because the reference fills a compound entity and the predictions leave it empty. These two tables are illustrative cases, not a prevalence estimate.

Fail-closed scoring keeps discarded failures in the mean, exposing the trade-off between output validity (Rule 112/120) and semantic chain correctness (SFT-1 0.159). We do not report a matched valid-only rescoring of the same predictions, so we do not claim that fail-closed scoring changes system ranking.

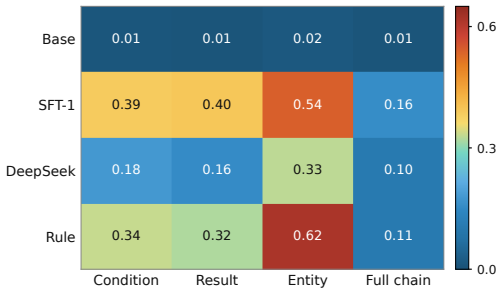


Figure 4: Field-level vs. chain-level F1 on Final Unseen (N=120). Rule entity F1 (0.617) can exceed SFT-1 while Rule chain F1 remains lower (0.106 vs. 0.159).

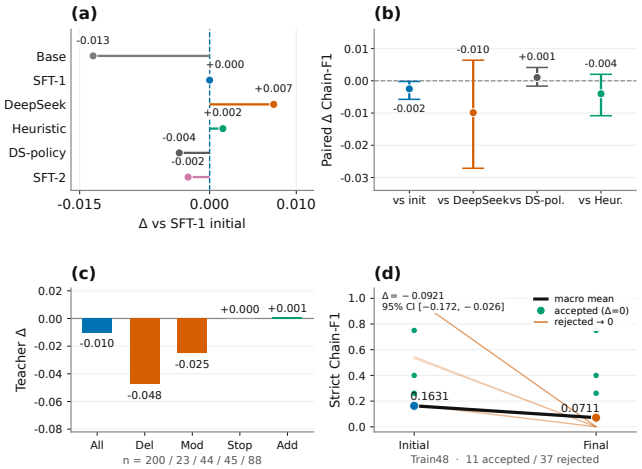


Figure 5: Source-only repair diagnostics. (a) Dev200 (N=200) Δ vs. frozen SFT-1. (b) SFT-2 paired 95% CIs; the vs.-initial interval is negative. (c) Teacher-action utility, including All n=200 (Δ=−0.0105); STOP yields Δ=0 by construction. (d) Train48 mean 0.163 → 0.071 (11 accepted / 37 rejected). Tested repair routes do not retain measurable utility once the reference is frozen.

5.5 RQ4: Do the tested learned repair routes add utility in their matched diagnostics?

Repair is evaluated on Dev200 and Train48 under the same source-only contract, never mixed into Table 6. Figure 5a plots Δ versus frozen SFT-1, including Base whole-table (0.141, Δ−0.014). DeepSeek whole-table has the highest observed Dev mean (0.162, Δ+0.007),

Table 7: Reference sensitivity: the same frozen Unseen predictions scored against isolated calls A and B and the adjudicated final set, not three independent gold standards. Bold is the highest Chain-F1 in each row. Δ is paired Chain-F1; 0∈CI marks a source-clustered 95% interval that includes 0.

Ref.	Base	SFT-1	DS	Rule	SFT-1−DS	DS−Rule
A	0.010	**0.147**	0.099	0.096	+0.049 (0∈CI)	+0.003 (0∈CI)
B	0.010	**0.181**	0.106	0.112	+0.075	−0.006 (0∈CI)
Final	0.008	**0.159**	0.105	0.106	+0.055 (0∈CI)	−0.002 (0∈CI)

Table 8: Frozen SFT-1 on Final Unseen, stratified by reference-chain count and by input-length tertile (boundaries 754 / 999 tokens). Exact match is Chain-F1 = 1. Valid denotes schema-compliant outputs.

Bin	N	Chain-F1	Cond. F1	Exact	Valid
Ref. chains 1–4	15	0.093	0.213	1	11
Ref. chains 5–8	32	0.140	0.406	1	28
Ref. chains 9–16	73	0.182	0.414	10	61
Tokens short	40	0.303	0.537	9	34
Tokens mid	40	0.142	0.370	3	36
Tokens long	40	0.033	0.252	0	30

Table 9: Illustrative scores on Unseen table 114 (14 chains, accepted from annotator A). The rule recovers every entity ($F1$=1.0) but splits rows into per-cell chains, so strict Chain-F1 is 0.

Role	One chain / table-level score
Frozen reference	`condition`: Patient:1|Sex:Female|Age:41|Employment status:Sick leave|Relationship status:Married; `result`: Children (age):2 (11, 13); `entity`: {}
Frozen SFT-1	Same 14-chain set. Chain-F1 1.0; entity F1 1.0.
Deterministic rule	`condition`: Patient:1; `result`: Sex:Female (then Age, Employment, …). Chain-F1 0; entity F1 1.0.

without a paired CI. SFT-2 versus initial is Δ−0.0025 (95% CI negative: −0.0057 to −0.0002). SFT-2 schema-valid rate is 0.965 and action exact-match is 0.61: errors are not due to unparseable outputs.

Figure 5c: teacher post-action Chain-F1 is 0.144 vs. initial 0.154 (overall Δ−0.0105). DELETE (n=23) and MODIFY (n=44) have mean Δ−0.048 and −0.025. Historical closed-loop gains involved reference-derived selection, so they are not identifiable as source-only repair gains; under the reference-blind protocol, the tested routes do not retain measurable utility.

Figure 5d (Train48, train-only): mean Chain-F1 falls from 0.163 to 0.071 (Δ−0.092, CI [−0.172, −0.026]); 11 accepted (Δ=0) and 37 rejected to 0. PPO was not advanced to Final Unseen because SFT-2 failed the Dev200 utility gate and the teacher targets already reduced chain quality. Imitating those targets (action exact-match 0.61) is not independent utility when the teacher itself degrades the chain.

6 Discussion

The protocol changes the evaluation target. Document-level isolation, frozen references, and fail-closed Chain-F1 together measure source-only agreement on new documents, including system failures. Legacy R-rule on an overlapping split measures a different estimand. BioTRACE does not raise extraction quality; it raises the validity of the judgment about which gains survive once overlap, oracle selection, and denominator rewriting are controlled. We cannot attribute the legacy/Unseen discrepancy to source isolation alone.

Table 10: Repair diagnostics under the source-only contract. Δ uses full-precision recorded scores. A paired CI is reported only for the pre-specified SFT-2 vs. initial gate. Teacher Δ=−0.0105; SFT-2 vs. initial CI is negative; PPO is not an Unseen system.

(a) Dev200 (N=200, AI-only). Bold is the highest observed Dev mean.

System	Chain-F1	Δ vs. SFT-1	95% CI
Base whole-table	0.1407	−0.0135	—
Fixed SFT-1 initial	0.1541	0	—
DeepSeek whole-table	**0.1615**	**+0.0074**	—
Heuristic + corrector	0.1557	+0.0015	—
DeepSeek policy + corrector	0.1506	−0.0035	—
SFT-2 + corrector	0.1517	−0.0025	[−0.0057, −0.0002]

(b) Teacher-action target utility on Dev200. Overall teacher Δ=−0.0105 (0.1437 vs. 0.1541).

Action	n	Teacher Δ	Improved / unchanged / degraded
All	200	−0.0105	—
DELETE	23	−0.0475	3 / 17 / 3
MODIFY	44	−0.0252	0 / 37 / 7
STOP	45	0.0000	0 / 45 / 0
ADD	88	**+0.0013**	8 / 67 / 13

(c) Train48 and PPO. Train-only N=48: 11 accepted / 37 rejected; 0.1631 → 0.0711; Δ−0.0921, CI [−0.1718, −0.0262]. PPO: the development-set utility criterion was not met; no Unseen PPO estimate.

Validity is not chain correctness. Rule outputs are usually contract-valid yet bind cells into the wrong chains (entity F1 0.617, chain F1 0.106). DeepSeek often emits fluent JSON that fails the entity vocabulary (75/120 zeros). SFT-1 is usually valid and still scores 0 on most tables because of mismatches, not missing JSON. Fail-closed scoring exposes that trade-off and prevents survivorship bias from discarded failures. Without a matched valid-only rescoring, we do not claim that fail-closed scoring changes ranking.

Binding is a distinct and consequential gap. Field F1 overstates joint relational correctness: a chain is usable for search or KG construction only when condition, result, and entities are bound together. Absolute Unseen Chain-F1 of 0.16 is agreement with a frozen AI reference, not evidence that chains are suitable for unsupervised

knowledge-base population. On this sample, Chain-F1 is lowest on the longest token tertile (Table 8); that association is descriptive.

Repair must be judged source-only. Historical closed-loop increments used reference-derived selection. Under the frozen protocol, teacher actions have negative mean utility and SFT-2 does not outperform its SFT-1 initialization. Matching a teacher (exact-match 0.61) is not additional chain quality.

6.1 Limitations

The Unseen reference is AI-only, with no human gold or IAA; annotators A and B are role-tagged calls of one model (mean A–B Chain-F1 0.494; 85/120 merge/revise), and the DeepSeek baseline shares that model family. N=120 prefers shorter valid tables. The 16-chain cap binds reference and prediction (36/120 at the cap). There is no downstream retrieval metric. SFT-1 uses one seed and the final checkpoint. Unseen contrasts were specified before label release but are not a public preregistration and are unadjusted for multiple comparisons.

7 Conclusion

BioTRACE is a source-isolated protocol and benchmark for extracting knowledge chains from scholarly tables. It jointly controls document overlap, reference-derived repair, and valid-only scoring, and it scores unordered condition–result–entity sets under a frozen AI-only reference. Under that estimand, supervised extraction raises agreement over a base model and a deterministic rule on new sources, while absolute chain quality remains low, field recovery overstates binding correctness, and the tested learned repair routes do not add measurable utility. Strict evaluation and transparent failure accounting should precede claims about agentic repair for table knowledge acquisition.

8 Ethical Considerations

Extracted records are not clinical advice and require independent human review before downstream use. Source tables may be licensed. Code: https://github.com/uiber3015/BioTRACE. DeepSeek-V3.2 is an annotation and baseline service, not ground truth.

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